

HW0 LA Review in R

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```
library('ggplot2')  
library('dplyr')
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library('car')
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

```
library('lmtest')
```

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

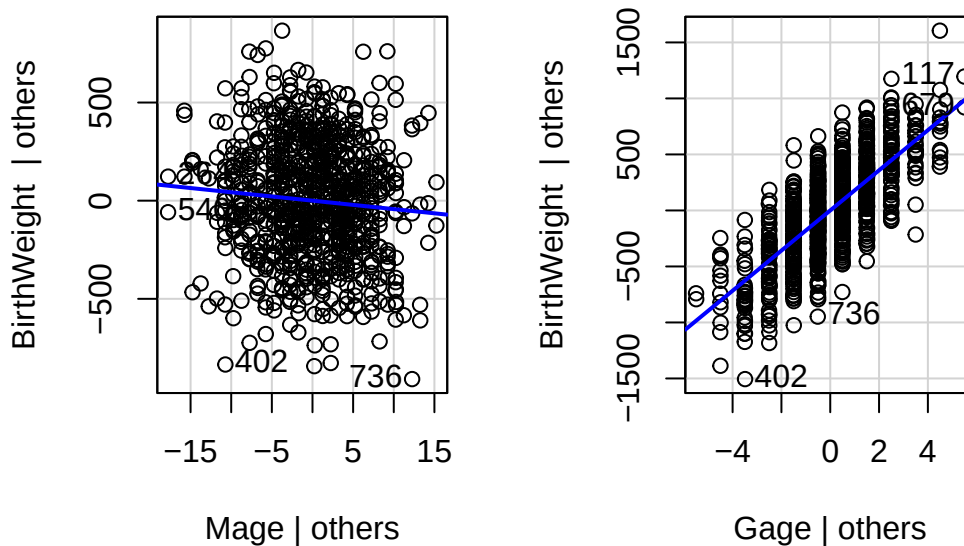
as.Date, as.Date.numeric

```
data <- read.table("BirthWeights.txt", sep = " ", header = TRUE, as.is = FALSE)
```

1.

```
mod <- lm(BirthWeight ~ Mage + Gage, data = data)
avPlots(mod)
```

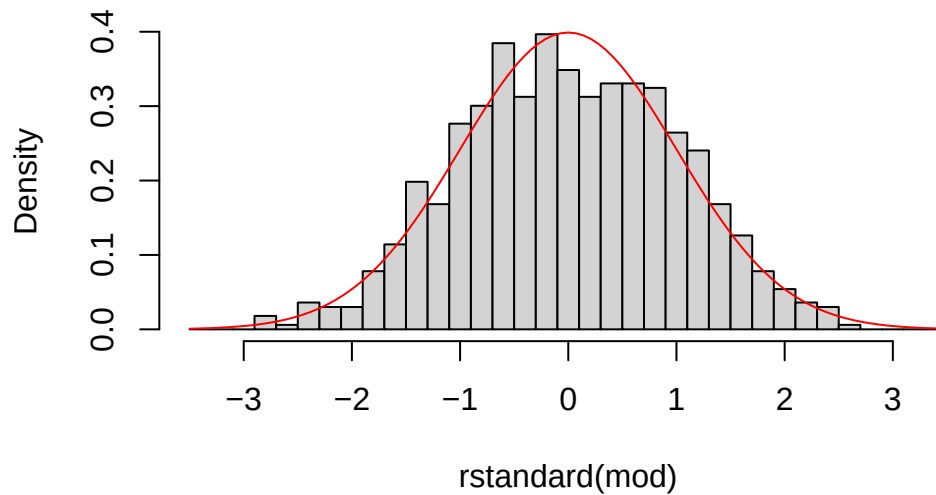
Added-Variable Plots



2.

```
mod <- lm(BirthWeight ~ ., data = data)
hist(rstandard(mod), freq = FALSE, breaks = seq(from = -3.5, to = 3.5, by = .2))
curve(dnorm(x), add = TRUE, col = 'red')
```

Histogram of rstandard(mod)



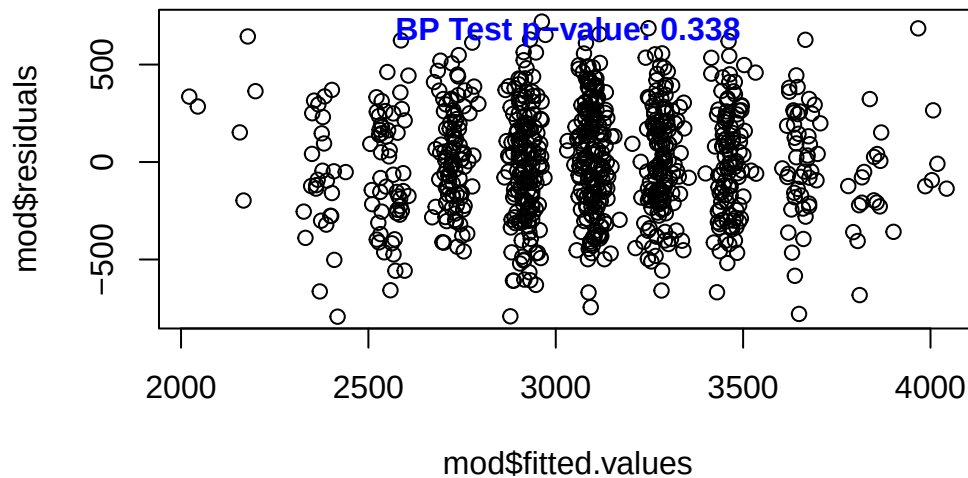
```
ks.test(rstandard(mod), "pnorm")
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data:  rstandard(mod)
D = 0.028947, p-value = 0.4884
alternative hypothesis: two-sided
```

3.

```
bp <- bptest(mod)
p_val <- bp$p.value
p_label <- paste("BP Test p-value:", format.pval(p_val, digits = 3))
plot(mod$fitted.values, mod$residuals)
mtext(p_label, side = 3, line = -1, font = 2, col = 'blue')
```



4.

```
summary(mod)$r.squared
```

```
[1] 0.6064689
```

5.

```
cross_validate <- function(fold_num){
  cv_data <- data[folds == fold_num, ]
  cv_mod <- lm(BirthWeight ~ ., data = cv_data)
  rmse <- sqrt(mean(cv_mod$residuals^2))
  rmse
}

set.seed(1337)
K <- 20
folds <- rep(1:K, length = nrow(data)) %>% #Rep K folds n times
  sample()
## Using sapply()
rmse_results <- sapply(1:K, FUN=cross_validate)

## Using lapply()
```

```
# rmse_results <- lapply(1:K, FUN=cross_validate) %>%
#   unlist()

## EVEN FASTER/BETTER/COOLER do the CV in parallel!
# library(parallel)
# rmse_results <- mclapply(1:K,FUN=cross_validate,mc.cores=detectCores()) %>%
#   unlist()
# Parallel does not work built-in in the current version of Positron

## Plot the Results
ggplot() + geom_histogram(mapping=aes(x=rmse_results))
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

